BASIS FOR DECISION MEMO

Permit Processor: Tom Braum
Date: October 2, 2023
Permit No. MI0020257
Designated Site Name: Cadillac WWTP

Monitoring Point 001A: Authorization to discharge 3.2 MGD of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River.

	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration						
Parameter	**Monthly**	**7-Day**	**Daily**	**Units**	**Monthly**	**7-Day**	**Daily**	**Units**	**Monitoring Frequency**	**Sample Type**	**Basis for Limits**
Flow	(report)	---	(report)	MGD	---	---	---	---	Daily	Report Total Daily Flow	PWJ
Carbonaceous Biochemical Oxygen Demand (CBOD5)											
June – September	190	270	(report)	lbs/day	7	---	10	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
October – November	290	**450**/*480*	(report)	lbs/day	11	---	17	mg/l	5x Weekly	24-Hr Composite	PWJ/AD
December – March	**610**/*670*	910	(report)	lbs/day	**23**/*25*	---	34	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April – May	450	670	(report)	lbs/day	17	---	25	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
Total Suspended Solids (TSS)											
June – September	530	800	(report)	lbs/day	20	30	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
October – May	800	1,200	(report)	lbs/day	30	45	(report)	mg/l	5x Weekly	24-Hr Composite	PWJ
Ammonia Nitrogen (as N)											
Through September 30, 2026											
June – September	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite	WQBEL
October – November	**43**/*88*	**75**	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite	PWJ/AD
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April – May	130	---	(report)	lbs/day	4.8	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
Beginning October 1, 2026											
June – September	24	53	(report)	lbs/day	0.9	---	2.0	mg/l	5x Weekly	24-Hr Composite	WQBEL
October – November	43	75	(report)	lbs/day	1.6	---	2.8	mg/l	5x Weekly	24-Hr Composite	PWJ/AD
December – March	99	---	(report)	lbs/day	3.7	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL/AD
April – May	**80**/*130*	---	(report)	lbs/day	**3.0**/*4.8*	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL

	Maximum Limits for Quantity or Loading				Maximum Limits for Quality or Concentration						
Parameter	**Monthly**	**7-Day**	**Daily**	**Units**	**Monthly**	**7-Day**	**Daily**	**Units**	**Monitoring Frequency**	**Sample Type**	**Basis for Limits**
Total Phosphorus (as P)											
May – March	13	---	(report)	lbs/day	0.5	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
April	17	---	(report)	lbs/day	0.65	---	(report)	mg/l	5x Weekly	24-Hr Composite	WQBEL
Total Copper	*0.3*	---	*0.82*	*lbs/day*	*11*	---	**(report)**/ *31*	ug/l	**2x Monthly**/ *Weekly*	24-Hr Composite	WQC
Total Nickel	*(report)*	---	*(report)*	*lbs/day*	*(report)*	---	*(report)*	*ug/l*	*Quarterly*	*24-Hr Composite*	
Total Selenium	*0.15*	---	*(report)*	*lbs/day*	*5.7*	---	*(report)*	*ug/l*	*Monthly*	*24-Hr Composite*	
Bis(2-ethylhexyl)phthalate	*0.82*	---	*(report)*	*lbs/day*	*31*	---	*(report)*	*ug/l*	*Monthly*	*24-Hr Composite*	
Total Arsenic	**0.43**	**---**	**(report)**	**lbs/day**	**16**	**---**	**(report)**	**ug/l**	**2x Monthly**	**24-Hr Composite**	**WQBEL**
Total Beryllium	**0.11**	**---**	**(report)**	**lbs/day**	**4.1**	**---**	**(report)**	**ug/l**	**2x Monthly**	**24-Hr Composite**	**WQBEL**
Total Cadmium	**0.11**	**0.32**	**(report)**	**lbs/day**	**4.1**	**---**	**12**	**ug/l**	**2x Monthly**	**24-Hr Composite**	**WQBEL**
Total Silver	**0.037**	**---**	**(report)**	**lbs/day**	**1.4**	**---**	**(report)**	**ug/l**	**2x Monthly**	**24-Hr Composite**	**WQBEL**
Chloride	**---**	**---**	**---**	**---**	**---**	**---**	**(report)**	**mg/l**	**Monthly**	**24-Hr Composite**	**WQC**
Sulfate	**---**	**---**	**---**	**---**	**---**	**---**	**(report)**	**mg/l**	**Monthly**	**24-Hr Composite**	**WQC**
Acute Toxicity – Fathead Minnow	**---**	**---**	**---**	**---**	**---**	**---**	**(report)**	**TU_A**	**Quarterly**	**24-Hr Composite**	**WQBEL**
Acute Toxicity – *C. dubia*	**---**	**---**	**---**	**---**	**---**	**---**	**(report)**	**TU_A**	**Quarterly**	**24-Hr Composite**	**WQBEL**
							Individual Chronic Value				
Chronic Toxicity – Fathead Minnow	**---**	**---**	**---**	**---**	**1.1**	**---**	**(report)**	**TU_C**	**Quarterly**	**24-Hr Composite**	**WQBEL**
Chronic Toxicity – *C. dubia*	**---**	**---**	**---**	**---**	**1.1**	**---**	**(report)**	**TU_C**	**Quarterly**	**24-Hr Composite**	**WQBEL**
							Maximum Daily				
Fecal Coliform Bacteria	---	---	---	---	200	400	(report)	cts/ 100 ml	5x Weekly	Grab	WQS
Perfluorooctanesulfonic Acid (PFOS)	*(report)*	---	*(report)*	*lbs/day*	*(report)*	---	(report)	ng/l	**3x Annual**/ *2x Annually*	Grab	WQC
Perfluorooctanoic acid (PFOA)	*(report)*	---	*(report)*	*lbs/day*	*(report)*	---	*(report)*	*ng/l*	*2x Annually*	*Grab*	
Available Cyanide	**0.16**	**---**	**(report)**	**lbs/day**	**5.9**	**---**	**(report)**	**ug/l**	**Monthly**	**Grab**	**WQBEL**

Parameter	Maximum Limits for Quantity or Loading Monthly	7-Day	Daily	Units	Maximum Limits for Quality or Concentration Monthly	7-Day	Daily	Units	Monitoring Frequency	Sample Type	Basis for Limits
Total Benzidine	**0.0032**	**---**	**(report)**	**lbs/day**	**0.12**	**---**	**(report)**	**ug/l**	**2x Monthly**	**Grab**	**WQBEL**
Total Bromomethane	**---**	**---**	**---**	**---**	**---**	**---**	**(report)**	**ug/l**	**Quarterly**	**Grab**	**WQC**
Total Mercury											
Corrected	(report)	---	(report)	lbs/day	(report)	---	(report)	ng/l	Quarterly	Calculation	WQC
Uncorrected	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Duplicate	---	---	---	---	---	---	(report)	ng/l	Quarterly	Grab	WQC
Field Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
Laboratory Method Blank	---	---	---	---	---	---	(report)	ng/l	Quarterly	Preparation	WQC
	12-Month Rolling Avg				**12-Month Rolling Avg**						
Total Mercury	0.000053	---	---	lbs/day	2.0	---	---	ng/l	Quarterly	Calculation	WQV
					Minimum % Monthly		**Minimum % Daily**				
TSS Minimum % Removal											
October – May	---	---	---	---	85	---	(report)	%	Monthly	Calculation	STS
					Minimum Daily		**Maximum Daily**				
pH	---	---	---	---	6.5	---	9.0	S.U.	5x Weekly	Grab	WQS
Dissolved Oxygen											
May – November	---	---	---	---	6.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD
December – April	---	---	---	---	5.0	---	---	mg/l	5x Weekly	Grab	WQBEL/AD

Basis for Limits Key
WQBEL - Water Quality-Based Effluent Limit
STS - Secondary Treatment Standard
WQC - Water Quality Concern
WQS - Water Quality Standard
WQV - Water Quality Variance
PWJ - Permit Writer's Judgment
AD – Antidegradation

Limit Change Key
Normal Type = existing requirement - carried over from previous version of permit
Bold Type = new requirement - not in previous version of permit
Italic = deleted requirement - not carried over from previous version of permit

PERMIT CONDITIONS:

- Final Effluent Limitations, Monitoring Point 001A
- Quantification Levels and Analytical Methods for Selected Parameters
- Additional Monitoring Requirements
- Pollutant Minimization Program for Total Mercury
- Pollutant Minimization and Source Evaluation Program for Perfluorooctanesulfonic Acid (PFOS), Perfluorooctanoic Acid (PFOA), **and Perfluorobutanesulfonic Acid (PFBS)**
- Untreated or Partially Treated Sewage Discharge Reporting and Testing Requirements
- Facility Contact
- Monthly Operating Reports
- Asset Management
- Discharge Monitoring Report – Quality Assurance Study Program
- **Continuous Monitoring**
- **PFAS Data Reporting Requirements**
- Storm Water Pollution Prevention (not required)
- Michigan Industrial Pretreatment Program
- Residuals Management Program for Land Application of Biosolids

NOTES:
For information pertaining to conventional pollutants and the establishment of certain permit conditions, see the Department's document named "ConvWQBEL_CadillacWWTP_HaringTwpWWTP_23."

For information pertaining to the non-attainment status for certain designated uses of the receiving water and how it informed the establishment of certain permit conditions, see the Department's document named "WQBELBiologistFactSheet_MI0020257_10-17-2022."

Monitoring requirements for chloride and sulfate have been included in the draft permit in accordance with the Department's Chloride and Sulfate Water Quality Values Implementation Plan (michigan.gov).

Monitoring and reporting requirements for PFOS, PFOA, and PFBS have been added to the permit in accordance with the Municipal NPDES Permitting Strategy for PFOS and PFOA (March 2023).

Less stringent limits are recommended for CBOD5 during the entire year, Dissolved Oxygen during the entire year, and for Ammonia Nitrogen from October - March. Based on a review of the facility's existing effluent quality, Permits Section determined that it is appropriate to retain the current limits under antidegradation.

Storm water discharges from the facility are covered by a No Exposure Certificate (NEC).

The monthly average effluent limitations for CBOD5 during the December – March season have been revised. This revision is consistent with Permits Section's translation of recommended daily maximum CBOD5 limitations protective of the dissolved oxygen standard in the receiving water.

The final effluent limits for CBOD5 and Ammonia Nitrogen during the October – November season are based on Permit Writer's Judgement. These effluent limits were calculated during the previous permit cycle. Maximum daily limits were calculated based on an estimate of the 99th percentile of the distribution of daily effluent data and monthly average limits were calculated based on an estimate of the 95th percentile of the distribution of monthly average effluent data; however record of the calculations cannot be located. Based on review of the facility's existing effluent quality, the facility continues to be able to meet these limits, so Permits Section determined that it is appropriate to retain the current limits under antidegradation.

A new chronic toxicity criterion for Ammonia Nitrogen was adopted by the Department in 2019 and is being implemented in NPDES permits. As part of this implementation, revised final effluent limitations for ammonia nitrogen that take effect on October 1, 2026 have been added to the draft permit. The implementation strategy also allows the permittee to complete an evaluation to verify whether the facility can comply with the revised effluent limits at the facility's design flow. If the evaluation determines the limits cannot be met, a facility-specific schedule of compliance for the revised limits would be developed taking into consideration the life of the treatment plant and the financial capability of the permittee. The City of Cadillac has chosen to forego the option to complete this evaluation to certify whether the revised limits can be met, thus the limits will take effect on October 1, 2026. For information pertaining to the establishment of these requirements, see the Department's April 27, 2023 memo, "Factors for Water Resources Division Staff to Consider When Implementing the 2019 Quality Values for Ammonia Update."

Based on a recommendation from district staff, Part I.C.1. Michigan Industrial Pretreatment Program requires an evaluation of whether local limits need to be developed and submitted to the Department for toxic pollutants with new or revised final effluent limits or monitoring.

The Department had a virtual meeting with the permittee on August 29, 2023 to discuss the permittee's questions and comments regarding the proposed draft permit. During this call, the monitoring frequency and timeframe for requesting a monitoring frequency reduction for the newly added toxic parameters were discussed. After the meeting, WRD management determined that it would be acceptable to reduce the monitoring frequency for Total Arsenic, Total Beryllium, Total Cadmium, Total Silver, Total Copper, and Total Benzidine from weekly to twice monthly and to reduce the timeframe for which a monitoring frequency reduction could be requested from 24 months to 12 months. As part of the decision to make these changes, Part I.C.1.q. will require the permittee to conduct a survey of the facility's Significant Industrial Users (SIU) to determine the source of these pollutants.